

BERENS RIVER, MB, Canada

WMO: 711580

Lat: 52.3597N

Lon: 97.0219W

Elev: 222

StdP: 98.69

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.6	-31.8	-38.5	0.1	-34.5	-35.5	0.1	-31.7	9.6	-17.6	8.9	-18.3	1.4	270	0.667

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	27.3	19.8	25.8	19.0	24.3	18.3	21.4	25.6	20.3	24.2	19.3	23.0	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.9	15.0	24.0	18.8	14.0	22.7	17.8	13.1	21.7	63.5	25.6	59.5	24.3	56.1	22.9	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.2	7.2	-38.5	30.9	2.5	1.8	-40.3	32.2	-41.7	33.2	-43.2	34.2	-45.0	35.5		
			-38.6	23.3	2.5	1.6	-40.4	24.4	-41.9	25.4	-43.2	26.3	-45.0	27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-18.4	-16.5	-9.0	0.3	8.0	15.1	18.0	16.7	11.5	3.9	-5.7	-15.1
	DBStd	14.28	8.27	7.96	8.37	6.76	5.26	4.09	3.30	3.49	4.50	4.48	6.58	7.96
	HDD10.0	4098	881	742	590	293	103	7	0	1	36	197	472	777
	HDD18.3	6471	1139	975	848	540	322	111	48	73	208	449	722	1036
	CDD10.0	748	0	0	1	3	41	160	247	209	81	7	0	0
	CDD18.3	78	0	0	0	0	1	14	36	23	3	0	0	0
	CDH23.3	543	0	0	0	0	17	126	239	142	18	1	0	0
	CDH26.7	82	0	0	0	0	2	24	36	18	2	0	0	0
Wind	WSAvg	3.3	3.2	3.1	3.3	3.4	3.5	3.3	3.3	3.4	3.4	3.6	3.4	3.2
Precipitation	PrecAvg	467	16	11	20	18	44	67	56	62	65	38	28	17
	PrecMax	605	60	21	103	44	88	150	134	156	139	135	94	47
	PrecMin	175	0	3	1	0	0	17	1	0	9	1	1	0
	PrecStd	97	15	6	22	13	24	34	35	41	32	32	21	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.6	3.3	11.0	19.8	25.5	29.5	29.8	28.8	25.7	19.1	11.1	2.5
		MCWB	0.9	1.6	6.1	10.7	15.2	20.7	21.6	21.2	18.7	14.0	8.7	0.7
	2%	DB	-1.3	1.0	6.7	15.9	22.3	26.4	27.4	26.6	22.9	15.1	6.8	-0.2
		MCWB	-2.2	0.3	3.3	8.6	13.6	18.7	20.2	19.7	17.3	11.2	4.6	-1.1
	5%	DB	-3.9	-0.9	4.2	12.7	19.6	24.4	25.9	24.9	20.8	12.8	4.5	-1.8
		MCWB	-4.4	-1.8	1.8	6.8	12.2	17.5	19.5	18.8	16.1	9.9	2.9	-2.4
	10%	DB	-6.2	-4.2	2.0	9.9	16.9	22.6	24.4	23.3	18.9	10.9	2.6	-3.6
		MCWB	-6.7	-5.1	0.4	5.0	11.1	16.5	18.9	18.2	15.1	8.7	1.5	-4.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	2.1	6.9	11.3	16.8	22.7	23.3	22.1	19.9	14.4	9.1	0.7
		MCDB	1.5	2.7	11.0	18.7	22.4	28.7	27.8	26.8	23.9	18.7	10.8	2.0
	2%	WB	-2.0	0.4	3.7	8.9	14.9	20.1	21.7	21.0	18.2	12.0	5.0	-0.9
		MCDB	-1.5	1.0	6.3	14.4	19.7	24.4	25.6	25.2	21.7	14.5	6.4	-0.3
	5%	WB	-4.4	-1.8	2.0	7.2	13.2	18.6	20.5	20.0	16.7	10.2	3.0	-2.3
		MCDB	-3.9	-0.8	4.0	12.1	18.2	22.6	24.5	23.6	19.9	12.5	4.3	-1.8
	10%	WB	-6.6	-5.0	0.6	5.3	11.7	17.4	19.5	19.0	15.5	8.6	1.5	-4.1
		MCDB	-6.2	-4.3	2.0	9.2	16.1	21.5	23.3	22.3	18.7	10.8	2.6	-3.6
Mean Daily Temperature Range	5% DB	MDBR	9.1	10.8	11.5	11.4	11.3	10.8	10.3	10.1	9.5	7.3	7.1	8.4
		MCDBR	8.7	9.4	11.1	14.8	15.2	13.5	12.0	12.1	11.9	10.7	8.0	7.8
		MCWBR	8.3	8.4	8.2	9.0	8.4	7.4	6.7	6.8	7.3	7.3	6.5	7.3
	5% WB	MCDBR	8.9	8.6	9.9	14.1	12.6	11.1	10.6	10.5	10.4	9.5	7.6	7.9
		MCWBR	8.5	7.8	7.4	8.8	7.6	6.6	6.5	6.4	7.0	6.9	6.4	7.4
Clear-Sky Solar Irradiance	taub	0.261	0.259	0.305	0.345	0.354	0.374	0.387	0.364	0.332	0.296	0.261	0.238	
	taud	2.221	2.279	2.279	2.366	2.422	2.384	2.370	2.461	2.522	2.576	2.417	2.295	
	Ebn at Noon	737	870	887	891	898	878	859	863	856	829	755	716	
	Edh at Noon	65	83	103	109	108	113	113	98	82	63	54	50	
All-Sky Solar Radiation	RadAvg	0.86	1.82	3.08	4.63	5.47	5.72	5.70	4.74	3.29	1.80	0.91	0.65	
	RadStd	0.09	0.15	0.31	0.41	0.40	0.41	0.30	0.33	0.36	0.20	0.10	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	-0.45	-0.54	N/A	N/A	N/A	N/A

Nomenclature: See separate page